

Non-MMT Reduced Cross-Wave Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved six-cell reduced evaluation prepared in `2026-07-24-13-10-16_non_mmt_reduced_evaluation_and_cross_wave_comparison.md`. The remote launcher completed successfully on `2026-07-24`, covering:

- `polished_dataset + setpoints`, forward and backward;
- `simplified_dataset + setpoints`, forward and backward;
- `polished_dataset + actual_values`, forward and backward.

The reduced comparison is an interim model-development decision surface. It does not replace the official full-matrix TE Curve Verification Pipeline decision, reopen `global`, or authorize a program-registry promotion.

Evaluation Method

The six selected-model reports were inspected together with their metrics, operator logs, measured-versus-predicted curve collages, and shape-gated reranker outputs. The decision follows the canonical multi-index curve-first policy and keeps these axes separate:

- raw error;
- mean-centered shape fidelity;
- offset behavior;
- harmonic and phase fidelity;
- robustness and shape-pass rate;
- visual curve evidence.

The setpoint matrices evaluated seven candidates per direction. The polished actual-values matrix evaluated nine candidates per direction, including the sparse-RCIM GRU and LSTM references.

Six-Cell Decision Matrix

Dataset and input mode	Direction	Recommended candidate
Polished setpoints	Fw	<code>polished_setpoints_periodic_gru_sequence_Fw</code>
Polished setpoints	Bw	<code>polished_setpoints_periodic_mlp_harmonic_Bw</code>
Simplified setpoints	Fw	<code>simplified_setpoints_periodic_mlp_harmonic_Fw</code>
Simplified setpoints	Bw	<code>simplified_setpoints_periodic_gru_sequence_Bw</code>
Polished actual values	Fw	<code>polished_actual_values_periodic_gru_sequence_Fw</code>
Polished actual values	Bw	<code>polished_actual_values_periodic_gru_sequence_Bw</code>

The periodic GRU is recommended in four of the six cells. The periodic harmonic MLP is recommended in polished-setpoint backward and simplified-setpoint forward. This preserves both an explicitly time-windowed branch and a non-windowed harmonic branch.

Axis leaders remain deliberately separate

- polished setpoints: Wave 4.1 leads raw error and offset, periodic GRU leads shape, and periodic harmonic MLP leads harmonic fidelity;
- simplified setpoints: the tree anchor leads raw error and offset but fails the shape gate, while periodic GRU and periodic harmonic MLP lead the shape-preserving axes;
- polished actual values: periodic GRU dominates both directions, except that periodic harmonic MLP leads forward harmonic fidelity and Wave 4.1 leads forward offset behavior.

Important Metric Findings

Polished Setpoints

Wave 4.1 MAE robust is the raw-error and offset leader in both directions, but it does not win the composite curve-first decision. Forward recommends the periodic GRU, while backward recommends the periodic harmonic MLP.

Simplified Setpoints

The tree anchor has the lowest raw MAE in both directions, but its shape-pass rates are only `0.340206` forward and `0.288660` backward. The curve collages show stepped predictions and loss of measured high-frequency content. Accordingly, the tree remains a raw-error and offset reference rather than a promotion candidate.

The tree artifacts were also serialized with scikit-learn `1.6.1` and loaded during this pass with scikit-learn `1.8.0`, which emitted `InconsistentVersionWarning`. Tree results therefore carry an additional reproducibility caution until replayed in a compatible environment.

Polished Actual Values

The periodic GRU is the recommended candidate in both directions. Backward is the clearest result: it leads every inspected axis, with raw MAE `0.001333 deg`, mean percentage error `2.624934%`, P95 percentage error `5.437606%`, and shape-pass rate `0.989362`.

The sparse-RCIM GRU and LSTM references win no cell. On backward actual-values evidence, sparse-RCIM GRU ranks third and sparse-RCIM LSTM ranks sixth. Their curve evidence follows the low-frequency trend but smooths or under-represents the measured high-frequency ripple. They remain useful exploratory references, not active leaders.

Visual Evidence Review

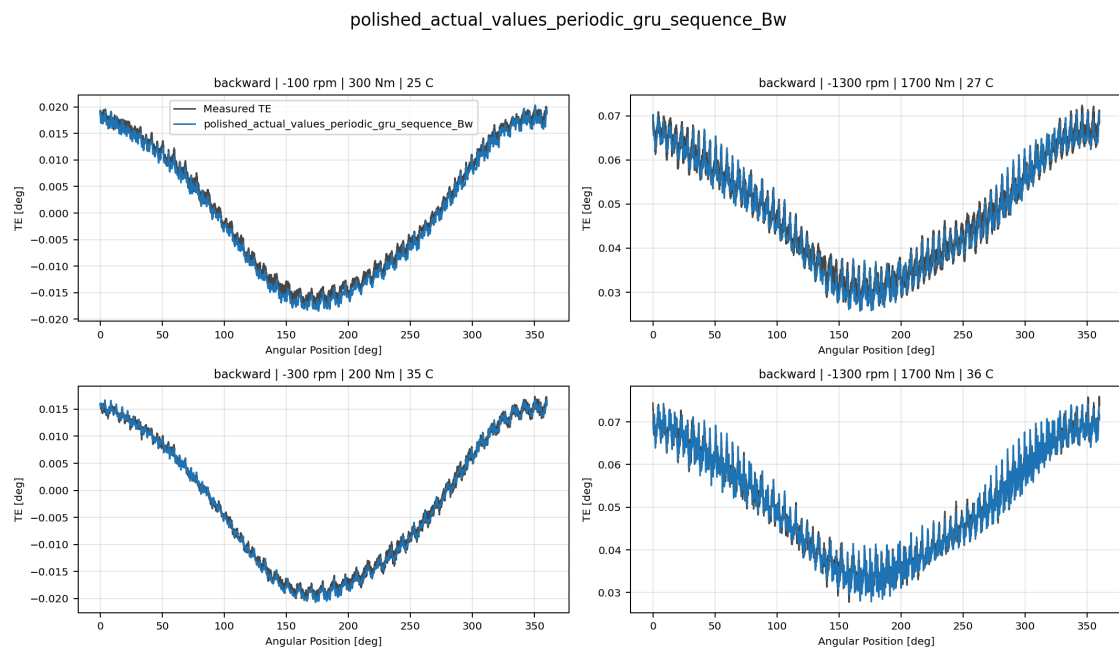
The six selected reports contain 46 candidate collages built from four shared conditions per direction. The visual review supports the metric decisions:

- periodic GRU preserves both the low-frequency TE envelope and measured high-frequency ripple most consistently on polished actual values;
- periodic harmonic MLP preserves useful harmonic structure on the two cells where it is recommended, although some offset mismatch remains visible;
- Wave 4.1 remains an important raw-error and offset diagnostic benchmark;
- the simplified-setpoint tree anchor looks stepped despite its scalar lead;
- sparse-RCIM temporal references smooth measured ripple and do not justify promotion.

Representative Curve Evidence

Polished Actual-Values Backward Winner

The periodic GRU follows both the low-frequency envelope and the measured high-frequency ripple across the shared operating conditions.

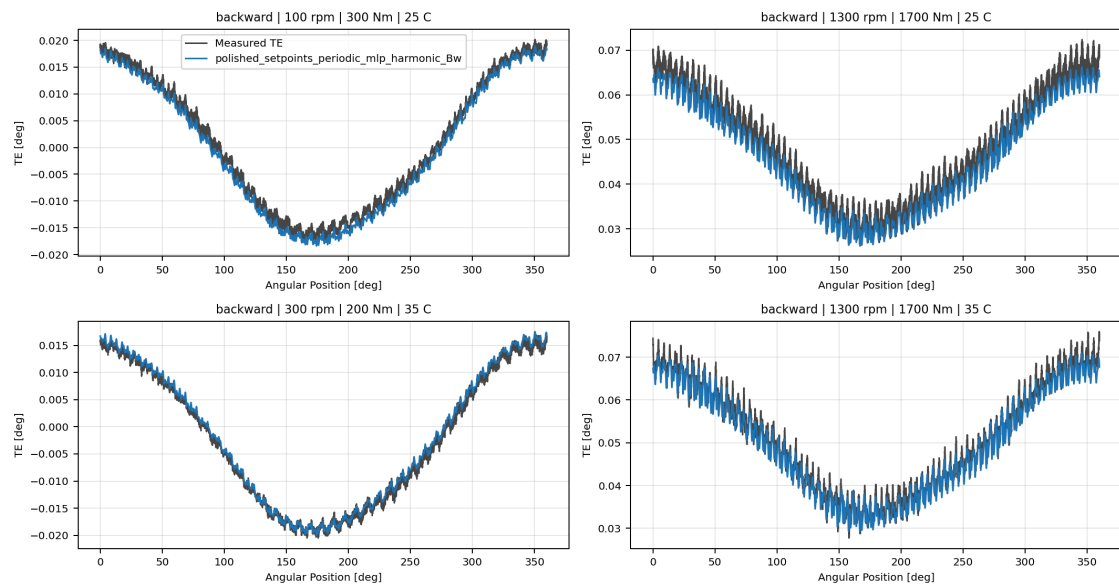


Polished actual-values backward periodic GRU curve evidence

Polished Setpoint Backward Winner

The periodic harmonic MLP retains the strongest composite harmonic and robustness balance for this cell, although local offset mismatch remains visible.

polished_setpoints_periodic_mlp_harmonic_Bw

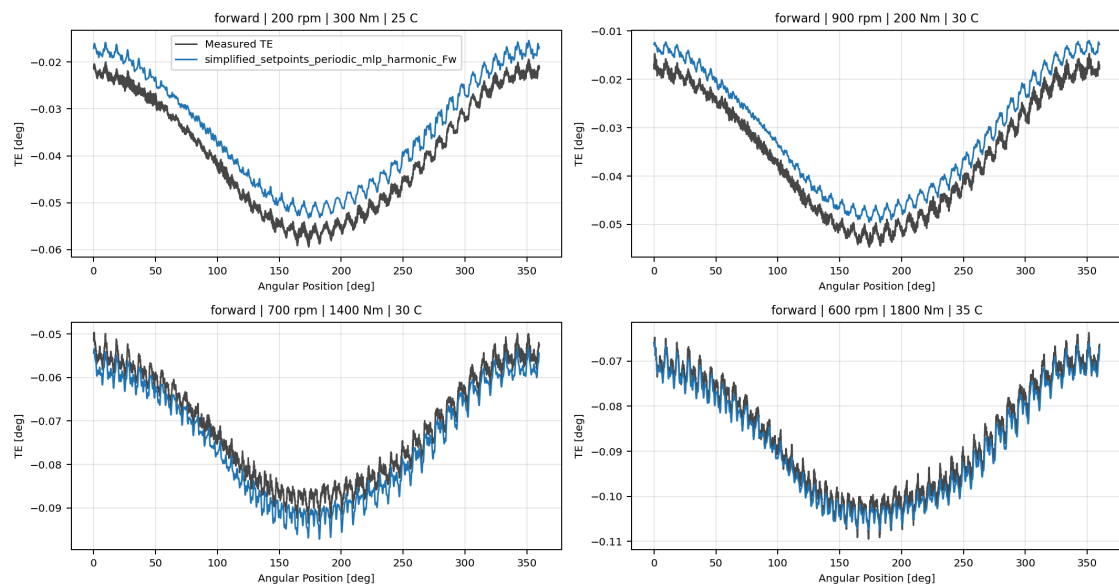


Polished setpoint backward periodic harmonic MLP curve evidence

Simplified Setpoint Forward Winner

The periodic harmonic MLP preserves the measured curve shape and ripple more faithfully than the scalar-leading tree anchor.

simplified_setpoints_periodic_mlp_harmonic_Fw

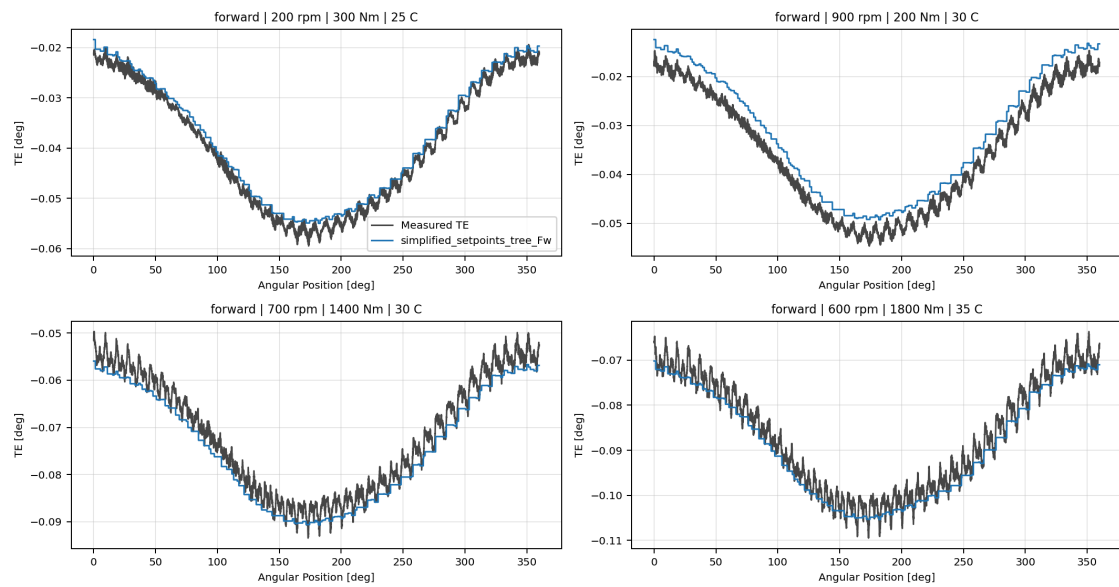


Simplified setpoint forward periodic harmonic MLP curve evidence

Simplified Tree Shape-Gate Caution

The tree prediction is visibly stepped and loses measured high-frequency content despite leading raw MAE and offset metrics.

simplified_setpoints_tree_Fw

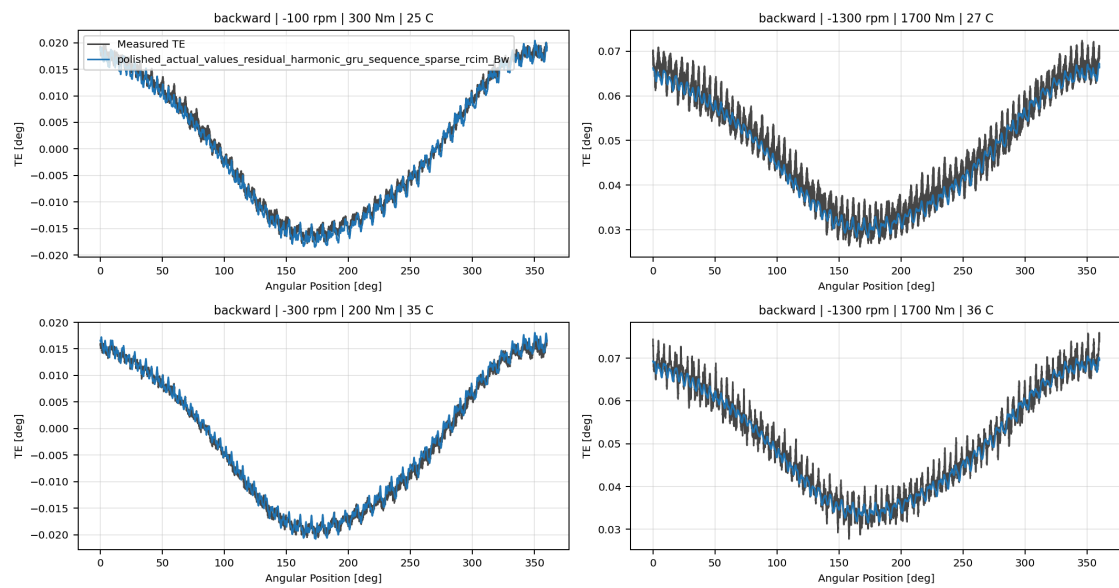


Simplified setpoint forward tree shape-gate caution

Sparse-RCIM Temporal Reference Caution

The sparse-RCIM GRU captures the broad low-frequency trend but smooths the measured ripple, supporting its reference-only status.

polished_actual_values_residual_harmonic_gru_sequence_sparse_rcim_Bw



Polished actual-values backward sparse-RCIM GRU caution

Program Decision

This reduced pass makes no official leader or registry change. The accepted official `Fw`, `Bw`, and `global` surfaces remain unchanged because this evaluation intentionally excluded the full official matrix and paused `global`.

The non-MMT active ingredients are now

1. `periodic_gru_sequence` as the primary time-windowed family;
2. `periodic_mlp_harmonic` as the required non-windowed harmonic complement;
3. `wave4_1_mae_robust_loss` as the raw-error and offset diagnostic ingredient;
4. `wave4_2_quantile_p10_p50_p90` as a secondary uncertainty benchmark.

Sparse-RCIM GRU and LSTM remain actual-values reference anchors only. MMT remains an inactive future TODO and is not required for the next branch.

Next Step

The reduced evidence now satisfies the previous prerequisite for Wave 6 planning. The next repository step is to prepare, but not train, a bounded technical design for a non-MMT Wave 6 integrated multi-head model. That design should combine the periodic GRU temporal path, the periodic harmonic MLP non-windowed path, and an explicit Wave 4.1-informed offset/raw-error head while preserving causal, TwinCAT-friendly intermediate quantities. Training remains gated behind a separate approved technical document and campaign plan.

Evidence Locations

- Selected reports:
`doc/reports/analysis/te_curve_verification_pipeline/04_selected_model_reports/[2026-07-24]/`
- Shape-gated reports:
`doc/reports/analysis/te_curve_verification_pipeline/03_cvp_diagnostics/non_mmt_cross_wave_shape_gated/`
- Remote operator logs:
`output/validation_checks/track2_operator_launch_logs/2026-07-24-13-30-33_reduced_selected_track2_reports/`
- Reduced matrix artifacts:
`output/validation_checks/track2_reference_comparison/`
- Shape-gated metric artifacts:
`output/validation_checks/non_mmt_cross_wave_shape_gated/`